

Foundational Research & Advances in Decoy-State QKD Protocol

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Decoy-State QKD Protocol in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Alice randomly sends bright pulses and dim pulses like fake decoys. If a spy tries to steal extra photons from multi-photon pulses, the math between the real and decoy pulses catches the thief."

3. Mathematical Formulation & Unitary Rule

$$Q_{\mu} = Y_0 + 1 - e^{-\eta \mu}, \quad Y_1 \geq \frac{\mu}{\mu \nu - \nu^2} \left[Q_{\nu} e^{\eta \nu} - \right]$$

Decoy state equations tightly bound the single-photon yield Y_1 and error rate e_1 .

5. Executable IBM Qiskit (Python) Code

```
import numpy as np

# Poissonian distribution for signal (mu=0.5) and decoy (nu=0.1)
mu, nu = 0.5, 0.1
p_single_signal = mu * np.exp(-mu)
p_single_decoy = nu * np.exp(-nu)
print(f"Single photon prob - Signal: {p_single_signal:.3f}, Decoy: {p_single_decoy:.3f}")
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Vulnerable to classical optical eavesdropping without quantum state monitoring. Foundational lossy optical fibers up to 100km.

7. Key Applications & Future Research Directions

- Commercial telecommunication dark fiber QKD networks
- Satellite-to-ground quantum communications links
- Government and banking inter-datacenter secure key exchange

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Cryptography pipelines.